

Exercise 2: SQL Aggregates & Operators

1) SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

2) SELECT department, AVG(age)
AS avg_age
FROM students
GROUP BY department;

department	avg_age
IT	20.5
HR	22
Finance	23

3) SELECT department, COUNT(student_id) AS students_count
FROM students
GROUP BY department
~~HAVING > 1 student~~
HAVING COUNT(student_id) > 1;

department	student count
IT	2
Finance	2

4) SELECT *
FROM students
WHERE age BETWEEN 21 ~~OR~~ ^{AND} 23;

student id	name
2	Bob
3	Charlie
4	Diana

5) SELECT student_id,
name,
~~age~~ age,
department

FROM students
WHERE department IN ('IT', 'HR') AND age > 21

student_id	name
2	Bob
5	Eve

Section 02 : The Courses Table

06) SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

department	total_credits
IT	11

07) SELECT course_id,
course_name,
department,
credits
FROM courses
WHERE credits != 4;

course_id	course_name	department	credits
T104	Excel	Finance	2
105	statistics	HR	3

08)

SELECT course_id,
course_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
103	Data Science	4
102	Python	4
105	statistics	3

Section 03: The enrollments Table.

07) SELECT MAX(grade) AS ~~max~~^{max-grade},
MIN(grade) AS min-grade,
AVG(grade) AS avg-grade
FROM enrollments;

Max-grade	Min-grade	avg-grade
90	78	84.6

10) SELECT course_id, COUNT(*) AS enrollment-count
FROM enrollments
GROUP BY course_id;

course_id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Section 04: The salaries Table.

11) SELECT department, SUM(salary) AS total-salary
SUM(bonus) AS total-bonus

FROM salaries
GROUP BY department;

department	Total salary	total bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12) SELECT department, AVG(salary) AS avg-salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;

department	avg-salary
IT	61000

13)

SELECT employee - id,

name,

salary,

bonus

SUM(bonus) AND SUM(salary) AS total - compensation

WHERE (salary + bonus) > 60000;

employee_id	name	Salary	bonus	total - compensation
1	Tom	60000	5000	65000
3	Spike	70000	60000	76000
4	Tora	62000	35000	65000

Table.

Section 05 Table

14) SELECT department,

SUM(budget) AS total - budget

AVG(budget) AS avg - budget

FROM projects

GROUP BY department

HAVING AVG(budget) > 70000

department	total - budget	avg - budget
IT	2700000	135000

15) SELECT project - id,

project - name,

department,

budget

FROM projects

WHERE budget BETWEEN 50000 AND 120000

AND department != 'Marketing'